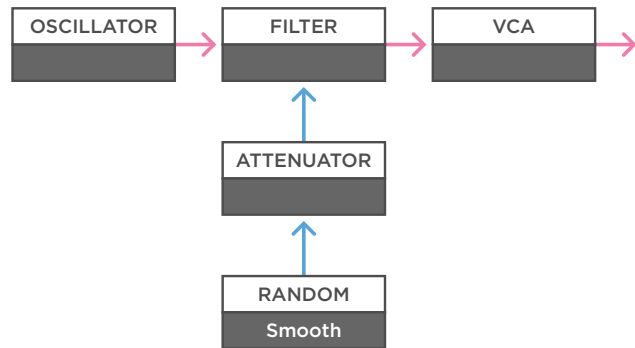


Smooth Random Voltages

Patch-tips

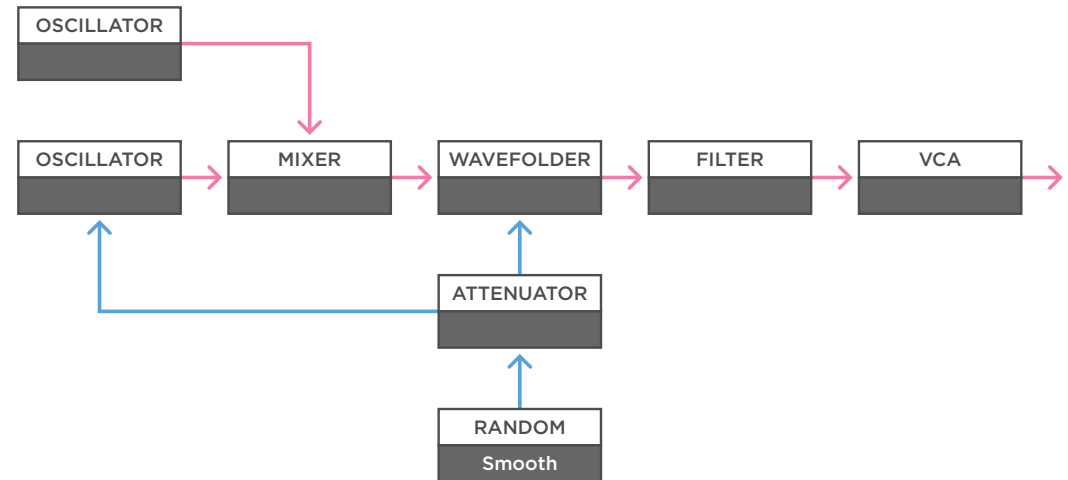
- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

a - In this example we use a straightforward monophonic voice with an oscillator, filter, and voltage-controlled amplifier. Then, we send a smooth random voltage through an attenuator to the filter. The attenuator is key, because although in some cases an extreme effect can be nice, most of the time an ever so slightly effect is enough. We don't want to create sounds that are impossible to control in a mix, we just want to add some light random texture.



(b)

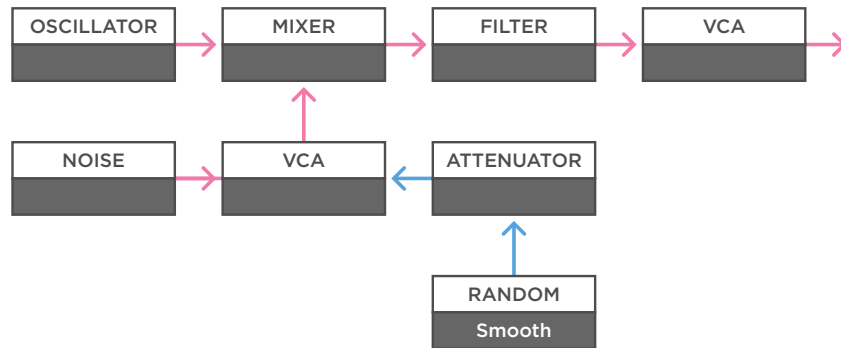
Other simple possibilities are sending the random voltage to the oscillator. This way you can create some controlled oscillator drifting. This can be especially nice if you use a mix of two oscillators in your patch, and only send the random voltage to one of them.

b - Here we mix two oscillators together. Tune one of them a fifth, seventh, or octave above, and feed the mix into a wavefolder. In this case a subtle random voltage to one of the oscillators or the wavefolder can have great results.

Smooth Random Voltages

Patch-tips

- Audio
- Control Voltage
- Trigger / Gate

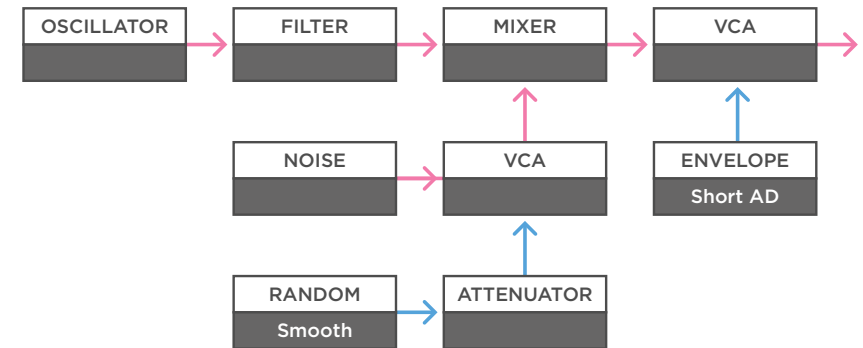


(a)

Using noise and mixing it with an oscillator before going to a filter is also a common setup. You can use the random voltage to modulate the filter and have some noise come through if the filter opens up. I find this needs a more opened filter to work properly though, and if you go a bit deeper in your sound-design you often want the filter and noise amount to be separately controllable.

a - So, a variation on this is to use a VCA between the noise source and the mixer. Then send the random voltage to the VCA, to control the amount of noise in the mix.

MONOTRAIL TECH TALK



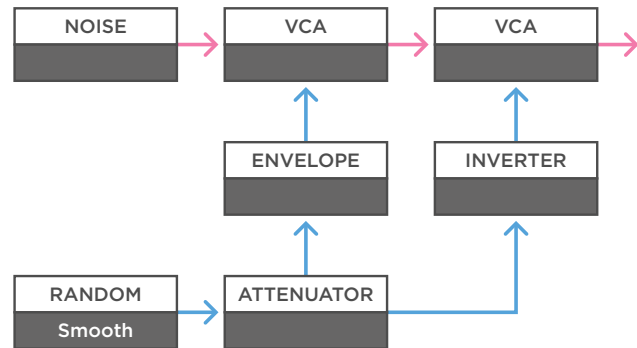
(b)

b - Sometimes it even works better to just pass the oscillator through the filter first, then use a mixer afterwards to add a random amount of unfiltered noise to the mix. This works well when you use a short attack decay envelope with a VCA at the end of the patch to create a more short and bright plucky or percussive sound. Although the amount of noise has some random to it, this way it's always bright and unfiltered.

Smooth Random Voltages

Patch-tips

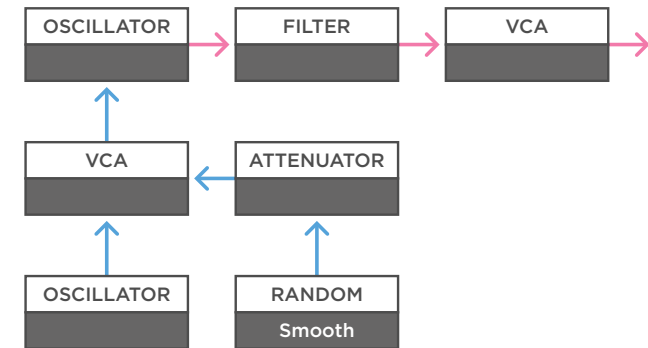
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Sending a random voltage to a VCA is a great way to add some dynamics to a steady synth or drum sound. If you make your own hats with a noise source, a VCA, and a voltage-controlled envelope for example. You can have the random voltage add some dynamics to the volume. You can also combine two tricks by multiplying the random voltage and control the decay of the envelope as well. If you invert the voltage to the VCA, you can create an inverted effect. This way, louder hats are short, and softer hats are longer.

MONOTRAIL TECH TALK



(b)

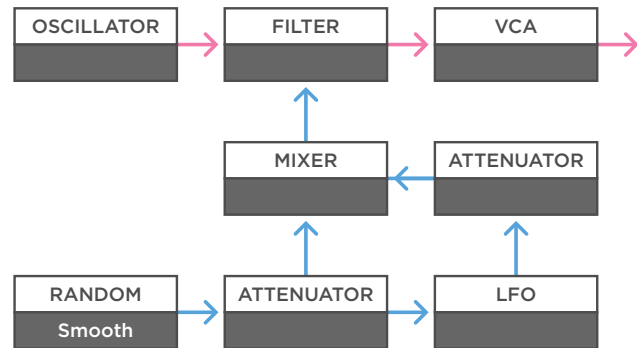
Subtle smooth random voltages to effect modules like the reverb amount or delay time can work great as well. And as long as the random is audible in the end result, it's worth it to experiment with more complex setups.

b - For example, use the random voltage to open a VCA that controls the amount of frequency modulation from one oscillator to another.

Smooth Random Voltages

Patch-tips

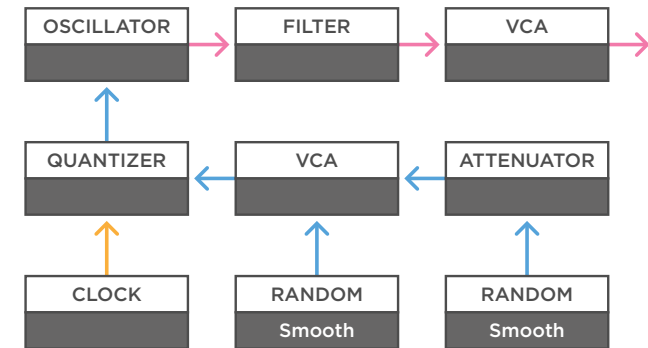
- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Or send the random voltage to influence the speed of an LFO that in turn modulates a filter, wavefolder or another parameter. You can expand the dynamics by sending one random voltage to multiple destinations. For example, take a random voltage to control both the speed of an LFO, as well as the filter. This way, when the LFO increases in speed, the filter opens the sound, creating a very natural effect.

MONOTRAIL TECH TALK



(b)

b - In this final setup we create a more complex random signal by using a slower random voltage to open a VCA that controls the amount of a faster random voltage passing through. You can use this complex random voltage on anything you like. For example, the amount of FM to an oscillator, or you can send the outcome to a quantizer to create a fast random melody, with a slowly changing maximum spread of notes.